

DISCRETE MATHEMATICS

Important Notes & Exam Revision Guide — Regulation 2025

Coverage: Mathematical Logic • Sets • Relations • Functions • Graph Theory • Combinatorics • Recurrence Relations

Exam focus: Definitions, standard laws, truth tables, properties, formulas, short proofs, and typical problem-solving methods.

UNIT I — MATHEMATICAL LOGIC

1. Propositions

- A proposition is a declarative statement that is either True (T) or False (F), but not both.
- Simple proposition: one statement. Compound proposition: formed using logical connectives.

2. Logical Connectives

- Negation: $\neg p$. Conjunction: $p \wedge q$. Disjunction: $p \vee q$. Exclusive OR: $p \oplus q$.
- Implication: $p \rightarrow q$ is false only when p is true and q is false.
- Biconditional: $p \leftrightarrow q$ is true when p and q have the same truth value.

3. Essential Logical Equivalences

- Double negation: $\neg(\neg p) \equiv p$.
- De Morgan: $\neg(p \wedge q) \equiv \neg p \vee \neg q$; $\neg(p \vee q) \equiv \neg p \wedge \neg q$.
- Implication: $p \rightarrow q \equiv \neg p \vee q$.
- Contrapositive: $p \rightarrow q \equiv \neg q \rightarrow \neg p$.
- Idempotent: $p \vee p \equiv p$; $p \wedge p \equiv p$. Commutative, associative and distributive laws also apply.

4. Tautology, Contradiction & Contingency

- Tautology: always true. Contradiction: always false. Contingency: sometimes true and sometimes false.
- To test equivalence, construct truth tables and compare the final columns.

5. Normal Forms

- CNF: conjunction (AND) of clauses, where each clause is a disjunction (OR) of literals.
- DNF: disjunction (OR) of terms, where each term is a conjunction (AND) of literals.
- Principal forms use every variable in each maxterm/minterm.

6. Predicates & Quantifiers

- Universal quantifier: $\forall x P(x)$ means $P(x)$ is true for every x in the domain.
- Existential quantifier: $\exists x P(x)$ means $P(x)$ is true for at least one x .
- Negation rules: $\neg \forall x P(x) \equiv \exists x \neg P(x)$; $\neg \exists x P(x) \equiv \forall x \neg P(x)$.

7. Rules of Inference

- Modus Ponens: $p, p \rightarrow q \Rightarrow q$. Modus Tollens: $\neg q, p \rightarrow q \Rightarrow \neg p$.
- Hypothetical syllogism: $p \rightarrow q, q \rightarrow r \Rightarrow p \rightarrow r$. Disjunctive syllogism: $p \vee q, \neg p \Rightarrow q$.

8. Proof Techniques

- Direct proof: assume the hypothesis and derive the conclusion.
- Contrapositive: prove $\neg q \rightarrow \neg p$ instead of $p \rightarrow q$.

- Contradiction: assume the statement is false and derive an impossibility.
- Mathematical induction: prove a base case, then prove $P(k) \rightarrow P(k+1)$.

UNIT II — SETS

1. Basic Concepts

- A set is a well-defined collection of distinct objects. Membership is denoted by $x \in A$.
- Subset: $A \subseteq B$ means every element of A belongs to B. Proper subset: $A \subset B$ and $A \neq B$.
- Empty set: $\emptyset$. Universal set: U. Power set: $P(A)$, the set of all subsets of A.

2. Operations

- Union: $A \cup B$. Intersection: $A \cap B$. Difference: $A - B$. Complement: A' or $U - A$.
- Useful laws: $A \cup \emptyset = A$, $A \cap U = A$, $A \cup A' = U$, $A \cap A' = \emptyset$.
- De Morgan: $(A \cup B)' = A' \cap B'$; $(A \cap B)' = A' \cup B'$.

3. Cardinality

- If $|A|=n$, then $|P(A)|=2^n$.
- For finite sets: $|A \cup B| = |A| + |B| - |A \cap B|$.
- For three sets: $|A \cup B \cup C|$ = sum of individual sizes – sum of pairwise intersections + triple intersection.

4. Cartesian Product

- $A \times B = \{(a,b): a \in A, b \in B\}$. If $|A|=m$ and $|B|=n$, then $|A \times B|=mn$.
- Generally $A \times B \neq B \times A$ unless special conditions hold.

UNIT III — RELATIONS & FUNCTIONS

1. Binary Relations

- A relation R from A to B is a subset of $A \times B$. A relation on A is a subset of $A \times A$.
- Reflexive: aRa for every a. Symmetric: $aRb \Rightarrow bRa$. Antisymmetric: aRb and $bRa \Rightarrow a=b$. Transitive: aRb and $bRc \Rightarrow aRc$.

2. Equivalence Relation

- A relation is an equivalence relation iff it is reflexive, symmetric and transitive.
- Equivalence relations partition a set into disjoint equivalence classes.

3. Partial Order

- A partial order is reflexive, antisymmetric and transitive.
- A partially ordered set (poset) is written $(A, \leq)$. Hasse diagrams are commonly used to represent finite posets.

4. Closure

- Reflexive closure adds (a,a) for missing a. Symmetric closure adds reverse pairs. Transitive closure adds pairs required for transitivity.
- Warshall's algorithm is a standard method for finding transitive closure from a relation matrix.

5. Functions

- A function $f:A \rightarrow B$ assigns exactly one element of B to each element of A.
- One-one (injective): $f(a_1)=f(a_2) \Rightarrow a_1=a_2$. Onto (surjective): every element of B has a preimage. Bijective: both one-one and onto.
- Inverse function f^{-1} exists as a function precisely when f is bijective.

- Composition: $(g \circ f)(x) = g(f(x))$.

UNIT IV — GRAPH THEORY

1. Basic Terminology

- A graph $G=(V,E)$ has vertices V and edges E . Degree of a vertex is the number of incident edges (with loops counted twice).
- Handshaking theorem: $\sum \deg(v) = 2|E|$. Therefore the number of odd-degree vertices is even.

2. Types of Graphs

- Simple graph: no loops or multiple edges. Multigraph: multiple edges may occur. Pseudograph may contain loops.
- Complete graph K_n has n vertices and $n(n-1)/2$ edges. Complete bipartite graph $K_{m,n}$ has mn edges.
- Regular graph: every vertex has the same degree. A k -regular graph has degree k at every vertex.

3. Walks, Paths & Connectivity

- Walk may repeat vertices/edges; trail has no repeated edge; path has no repeated vertex; circuit is a closed trail.
- A graph is connected if every pair of vertices is joined by a path.

4. Euler & Hamilton Graphs

- Euler circuit uses every edge exactly once and returns to the start. A connected graph has an Euler circuit iff every vertex has even degree.
- Euler trail uses every edge exactly once; a connected graph has one iff exactly 0 or 2 vertices have odd degree.
- Hamiltonian cycle visits every vertex exactly once and returns to the start.

5. Matrix Representation

- Adjacency matrix A has A_{ij} equal to the number of edges between v_i and v_j (for a simple graph, entries are 0 or 1).
- Incidence matrix records which vertices are incident with which edges.

6. Trees

- A tree is a connected acyclic graph. A tree with n vertices has $n-1$ edges.
- Every pair of vertices in a tree is connected by exactly one path.
- A spanning tree contains all vertices and is itself a tree.

UNIT V — COMBINATORICS

1. Counting Principles

- Rule of Sum: if choices are mutually exclusive, total = $m+n$.
- Rule of Product: if a process has m choices followed by n choices, total = mn .

2. Permutations

- $n! = n(n-1)\dots 2 \cdot 1$, with $0! = 1$.
- Number of arrangements of r objects from n distinct objects: $P(n,r) = n!/(n-r)!$.
- With repeated objects: $n!/(n_1! n_2! \dots n_k!)$.

3. Combinations

- $C(n,r) = n!/[r!(n-r)!]$.
- Symmetry: $C(n,r) = C(n,n-r)$. Pascal identity: $C(n,r) = C(n-1,r-1) + C(n-1,r)$.

4. Binomial Theorem

- $(x+y)^n = \sum_{r=0}^n C(n,r)x^{n-r}y^r$.
- General term: $T(r+1)=C(n,r)x^{n-r}y^r$.

5. Pigeonhole Principle

- If $n+1$ objects are placed into n boxes, at least one box contains at least two objects.
- Generalized form: if N objects are distributed among k boxes, some box contains at least $\text{ceil}(N/k)$ objects.

6. Inclusion-Exclusion

- For two sets: $|A \cup B| = |A| + |B| - |A \cap B|$.
- For three sets: add singles, subtract pairwise intersections, then add the triple intersection.

UNIT VI — RECURRENCE RELATIONS

1. Definition

- A recurrence relation defines a sequence using one or more earlier terms, together with initial conditions.
- Example: $a_n = a_{n-1} + a_{n-2}$, with suitable initial values.

2. First-Order Linear Recurrence

- For $a_n = c a_{n-1} + d$, solve the associated homogeneous part and then find a particular constant solution when applicable.
- Always use the initial condition to determine the arbitrary constant.

3. Second-Order Homogeneous Relation

- For $a_n = c_1 a_{n-1} + c_2 a_{n-2}$, form the characteristic equation $r^2 - c_1 r - c_2 = 0$.
- Distinct roots r_1, r_2 : $a_n = A r_1^n + B r_2^n$.
- Repeated root r : $a_n = (A + Bn)r^n$.

4. Non-Homogeneous Relations

- Find the complementary (homogeneous) solution first, then a particular solution, and add them.
- The trial form for the particular solution depends on the forcing term; if it overlaps with the complementary solution, multiply the trial by an appropriate power of n .

5. Generating Functions

- The ordinary generating function of $\{a_n\}$ is $A(x) = a_0 + a_1x + a_2x^2 + \dots$.
- Generating functions convert recurrence problems into algebraic equations that can be solved for $A(x)$, followed by coefficient extraction.

QUICK FORMULA SHEET

Topic	Key formulas / facts
Logic	$p \rightarrow q \equiv \neg p \vee q$; $\neg(p \wedge q) \equiv \neg p \vee \neg q$; $\neg(p \vee q) \equiv \neg p \wedge \neg q$
Sets	$ A \cup B = A + B - A \cap B $; $ P(A) = 2^{ A }$
Relations	Equivalence = reflexive + symmetric + transitive; Poset = reflexive + antisymmetric + transitive
Functions	Bijjective = injective + surjective; $(g \circ f)(x) = g(f(x))$
Graphs	$\sum \deg(v) = 2E$; K_n has $n(n-1)/2$ edges; tree with n vertices has $n-1$ edges
Counting	$P(n,r) = n!/(n-r)!$; $C(n,r) = n!/[r!(n-r)!]$
Recurrence	Distinct roots: $a_n = A r_1^n + B r_2^n$; repeated root: $a_n = (A + Bn)r^n$

EXAM REVISION CHECKLIST

- Practice truth tables and logical equivalence without referring to notes.
- Memorize the four key relation properties and identify them from ordered pairs or matrices.
- Practice one-one/onto/bijjective tests and inverse/composition problems.
- Memorize graph degree, Euler conditions, complete-graph edge counts, and tree properties.
- Practice permutation/combination selection: ask first whether order matters.
- For recurrence relations, write the characteristic equation carefully and apply initial conditions at the end.
- Before the exam, solve at least one problem from every listed subtopic.